GEOG 172

Describe Point Patterns

We will look at one type of spatial data and learn how to analyze it for departures from CSR

Lecture 4-1

POINT PATTERNS

A set of *n* events at locations $S = [s_1, s_2, \ldots, s_n]$ in which each event (or point) s_i has location coordinates x_i, y_i, which occurs in a study region *a*.

QUESTION

Does this definition remind you have something you learned in Week 1?

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ANSWER: This definition is very similar to the GeoAtom. This similarity is not surprising because a point pattern is a set defined by the location (and time) components of the GeoAtom. The marked point pattern brings back in the attributes.

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1. The pattern should be mapped on the plane
2. The study area should be objectively determined
3. All relevant entities in the study are should be included
4. Locations should represent entities that can be considered at the scale of the study area

TWO APPROACHES

Intensity-based Approaches: focused on assessing the concentration of events within space.

Separation-based Approaches: focused on assessing the distance between events

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Intensity-based Approaches: focused on assessing the concentration of events within space.

- First–Order Variation – Variation in some factor make it more/less likely for events to occur at particular locations.

Separation-based Approaches: focused on assessing the distance between events

- Second–Order Variation – Interactions between locations depend on the distance between those locations

THE RANDOM ORCHARD AUDIT

Were those trees randomly placed?

ACTIVITY

Overview: The Department of Random Agriculture has called for an audit of 8 orchard plots after receiving criticism that their evaluation techniques use only a density-based approach. To assess appease the public, auditors have been sent out to evaluate these orchards using an alternative separation-based approach.

> *"Your inspectors only looked at how many trees occur in different parts of the orchard. You never considered how far apart the trees actually are."*

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Purpose: Become familiar with separation-based approach to CSR assessment.

Procedure A1:

- With your team, familiarize yourselves with **Worksheet A**
- Identify candidate nearest neighbors, measure their distances (https://www.calculator.net/distance-calculator.html)
- Identify and record each nearest neighbor and the corresponding distance

ACTIVITY

Overview: Auditors have been sent out to evaluate these orchards using an alternative separation-based approach.

Procedure A2:

- Calculate the observed Nearest Neighbor Statistic
- Calculate the expected Nearest Neighbor Statistic
- Calculate the Nearest Neighbor Index

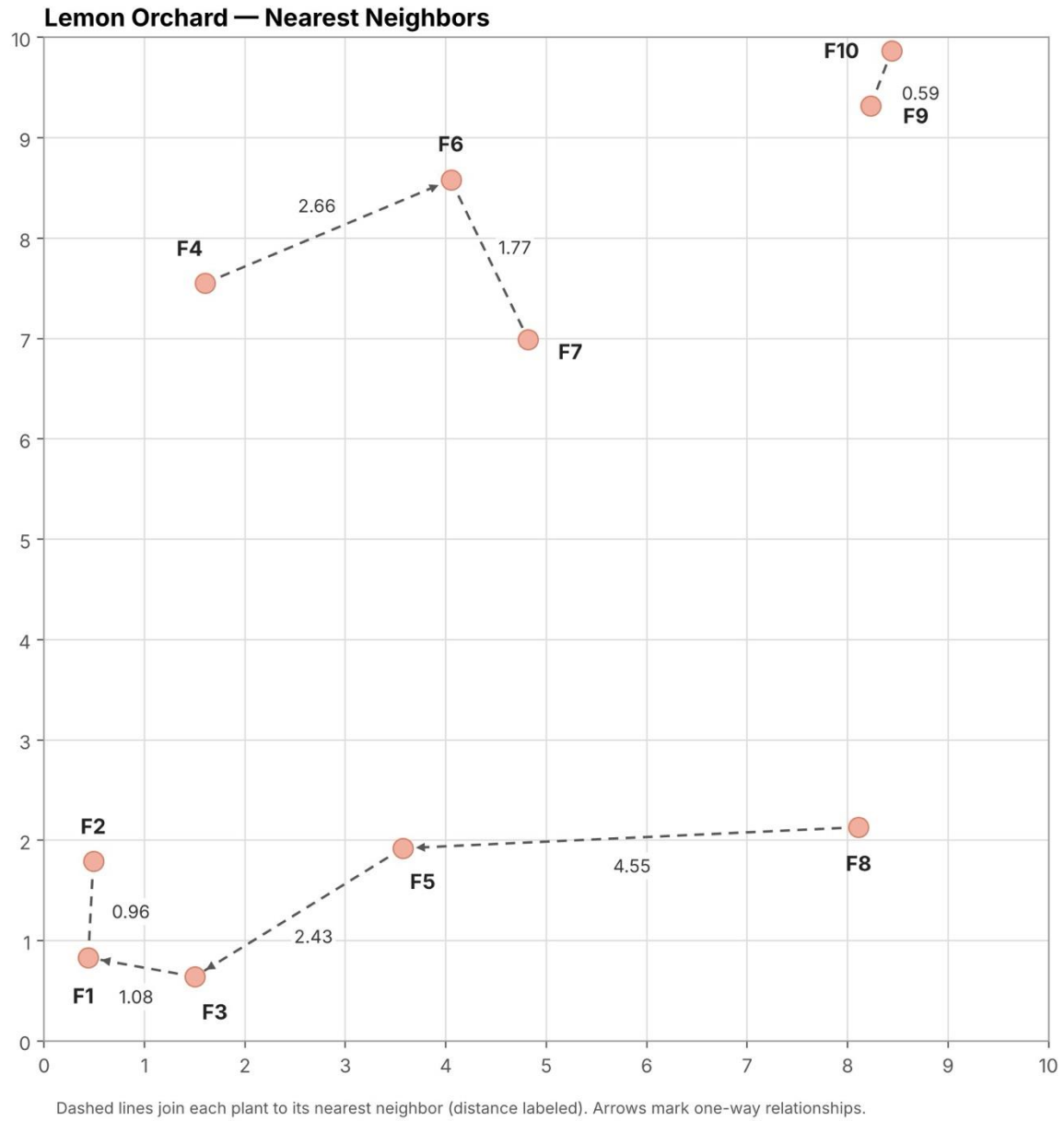
$$R = \frac{\overline{r_{obs}}}{\overline{r_{exp}}} = \frac{\frac{\sum_{i=1}^{n} d_i}{n}}{\frac{1}{2\sqrt{n/A}}}$$

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EDGE EFFECTS

Points near the boundary of the study region have less space to have neighbors. The true nearest neighbor may be outside the observed study area. The existence of edge effects can alter a range of spatial statistics

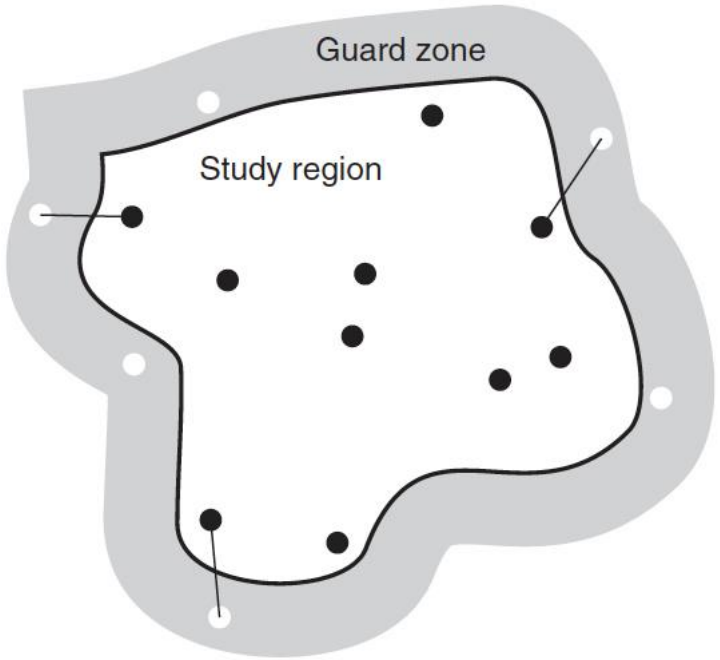


Figure 5.13 The use of a guard zone in a distance-based point pattern measure.

ACTIVITY

Overview: Auditors have been sent out to evaluate these orchards using an alternative separation-based approach. This audit will also account for *Edge Effects*.

Procedure B1:

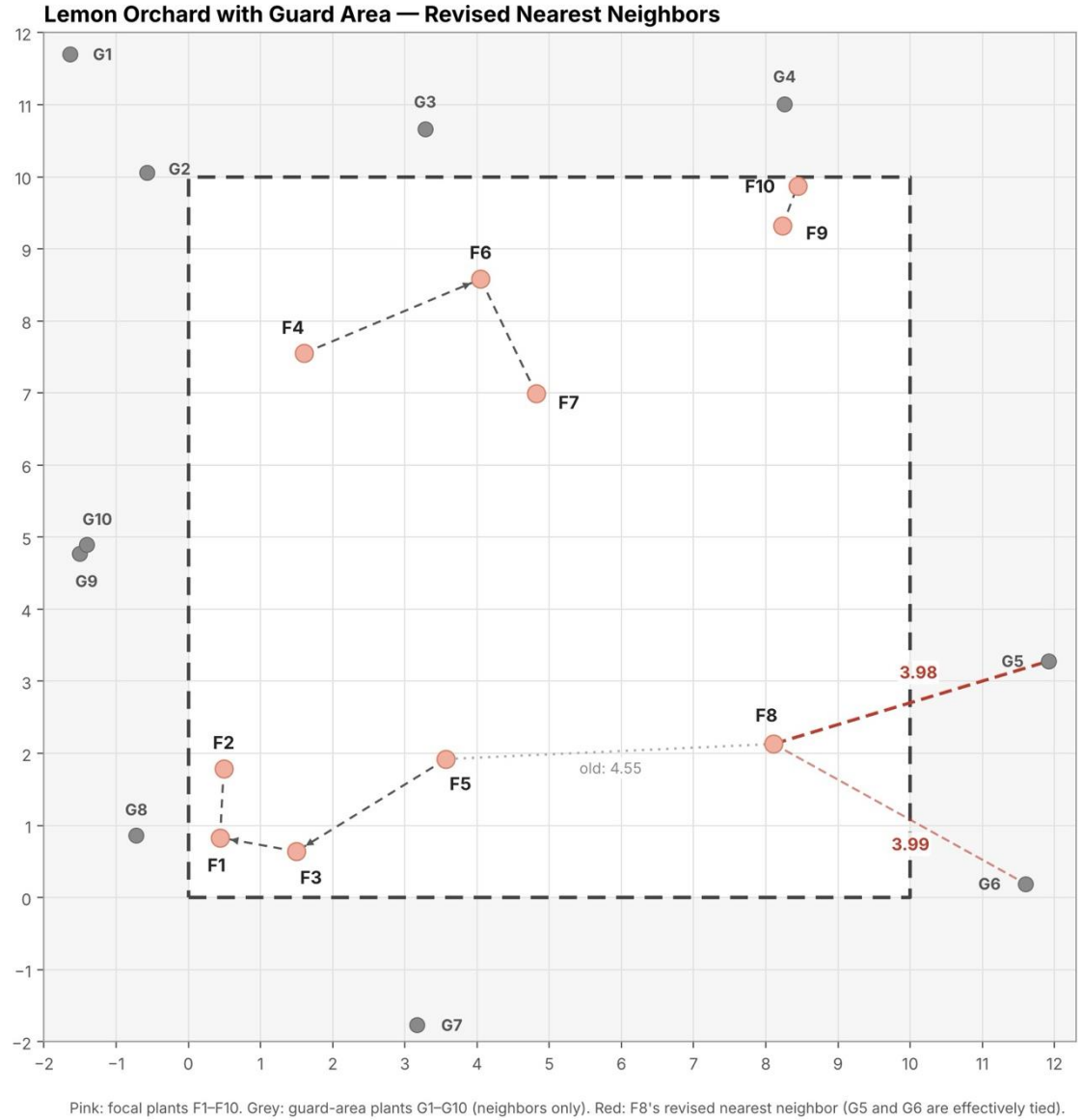
- Use Worksheet B to recalculate the Nearest Neighbor Statistic considering the guard area

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Procedure B1:

- Use Worksheet B to recalculate the Nearest Neighbor Statistic considering the guard area



ACTIVITY

Overview: Auditors have been sent out to evaluate these orchards using an alternative separation-based approach. This audit will also account for *Edge Effects*.

Procedure B1:

- Use Worksheet B to recalculate the Nearest Neighbor Statistic considering the guard area

QUESTION

Should neighbors in the Guard Area receive the same weight as neighbors in the study area?

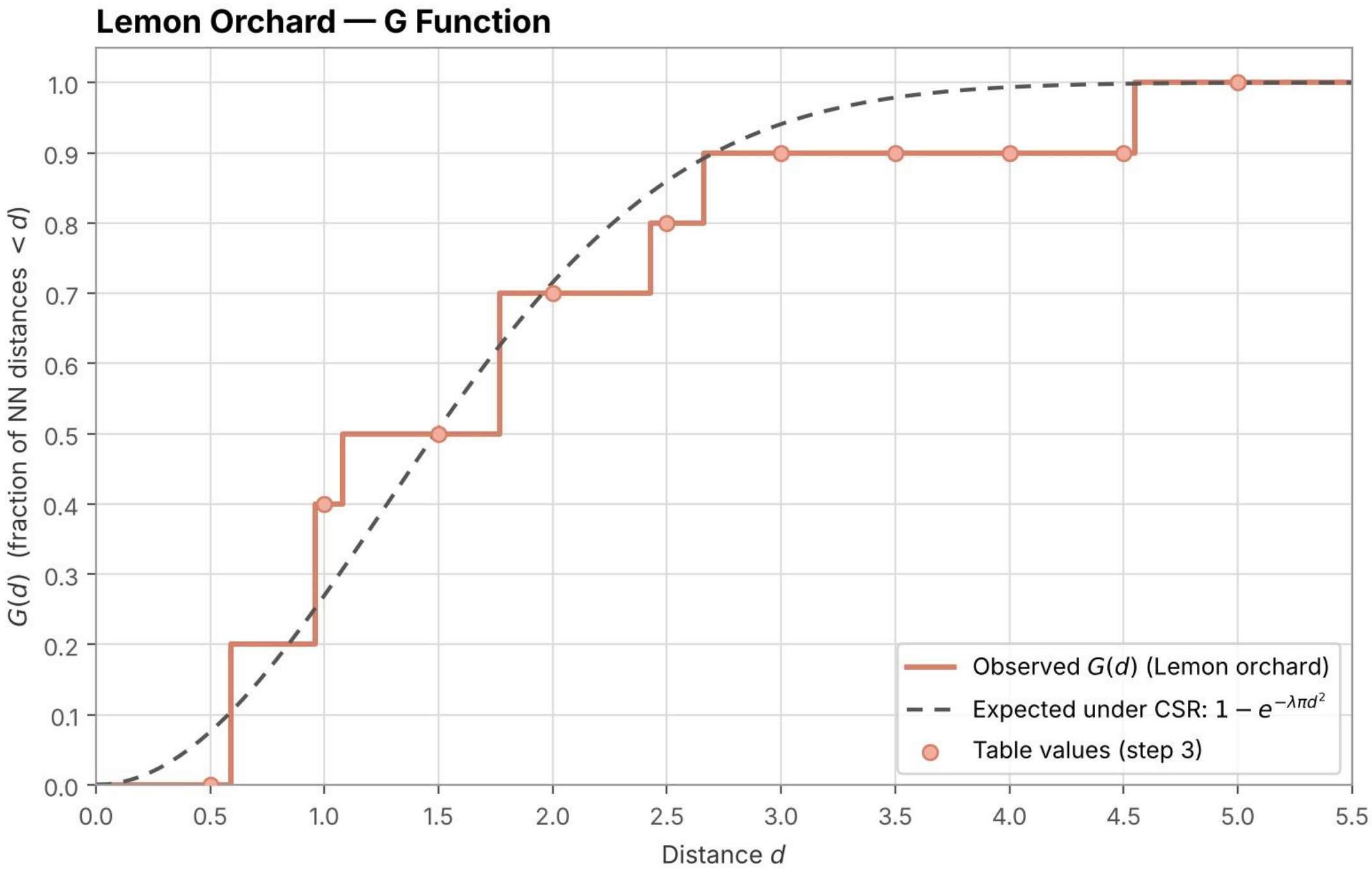
Extensions of Nearest Neighbor

There is more information to use

G FUNCTION

Examines the cumulative frequency distribution of the NN distances. For any distance, *G* provides the fraction of the NN distances less than *d*.

$$G(d) = \frac{Count[d_{min}(s_i) < d]}{n}$$

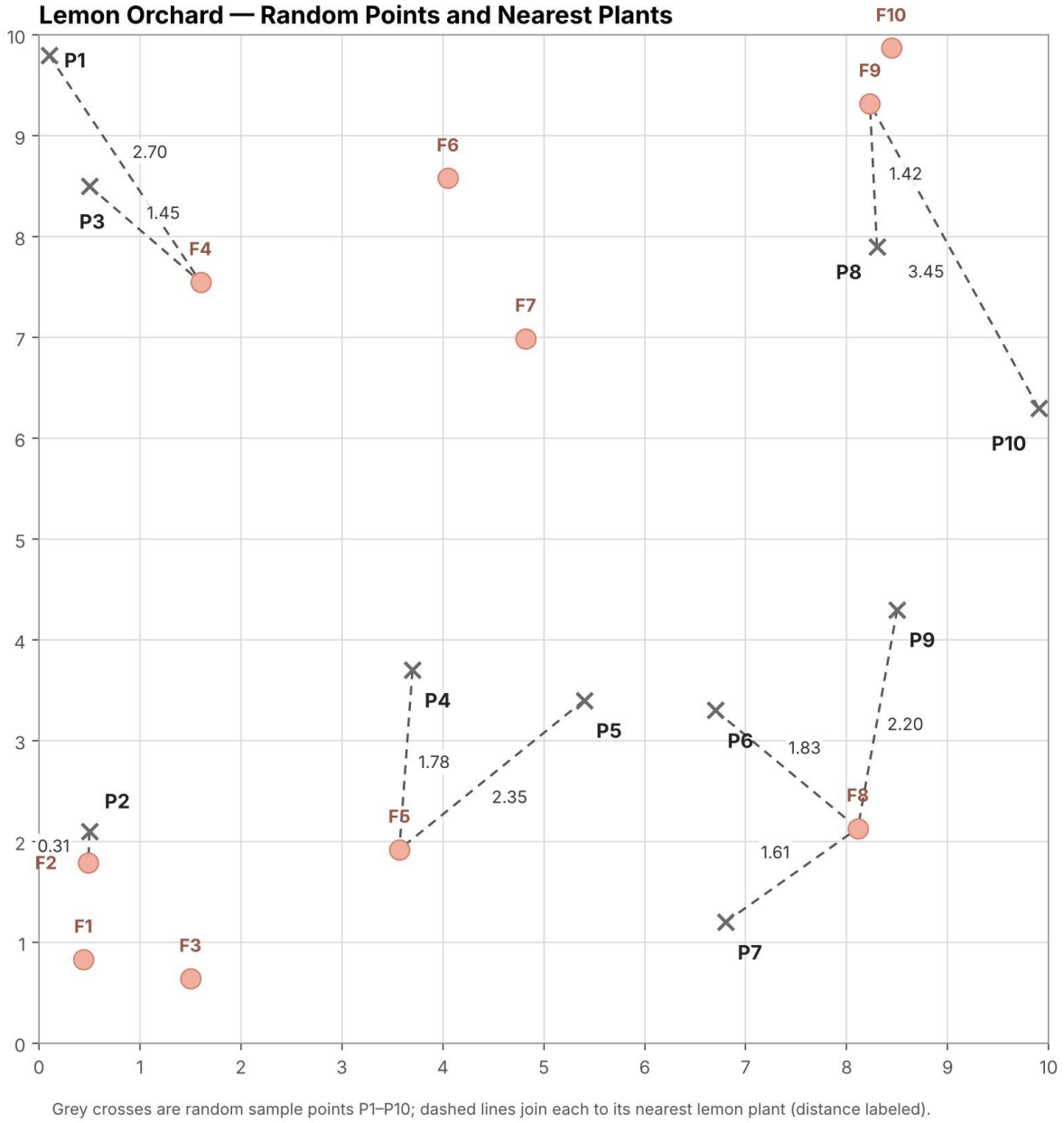

Lemon Orchard — G Function
G(d) (fraction of NN distances < d)
Distance d
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
5.5
Observed G(d) (Lemon orchard)
Expected under CSR: 1 − e^{−λπd²}
Table values (step 3)

F FUNCTION

Examines the cumulative frequency of NN distances from a set of points randomly placed in the study area to the closest point in the pattern

$$F(d) = \frac{Count[d_{min}(p_i, s_i) < d]}{m}$$

Where $d_{min}(p_i, s_i)$ is the minimum distance from location p_i in the randomly selected set to any event in the point pattern *S*.

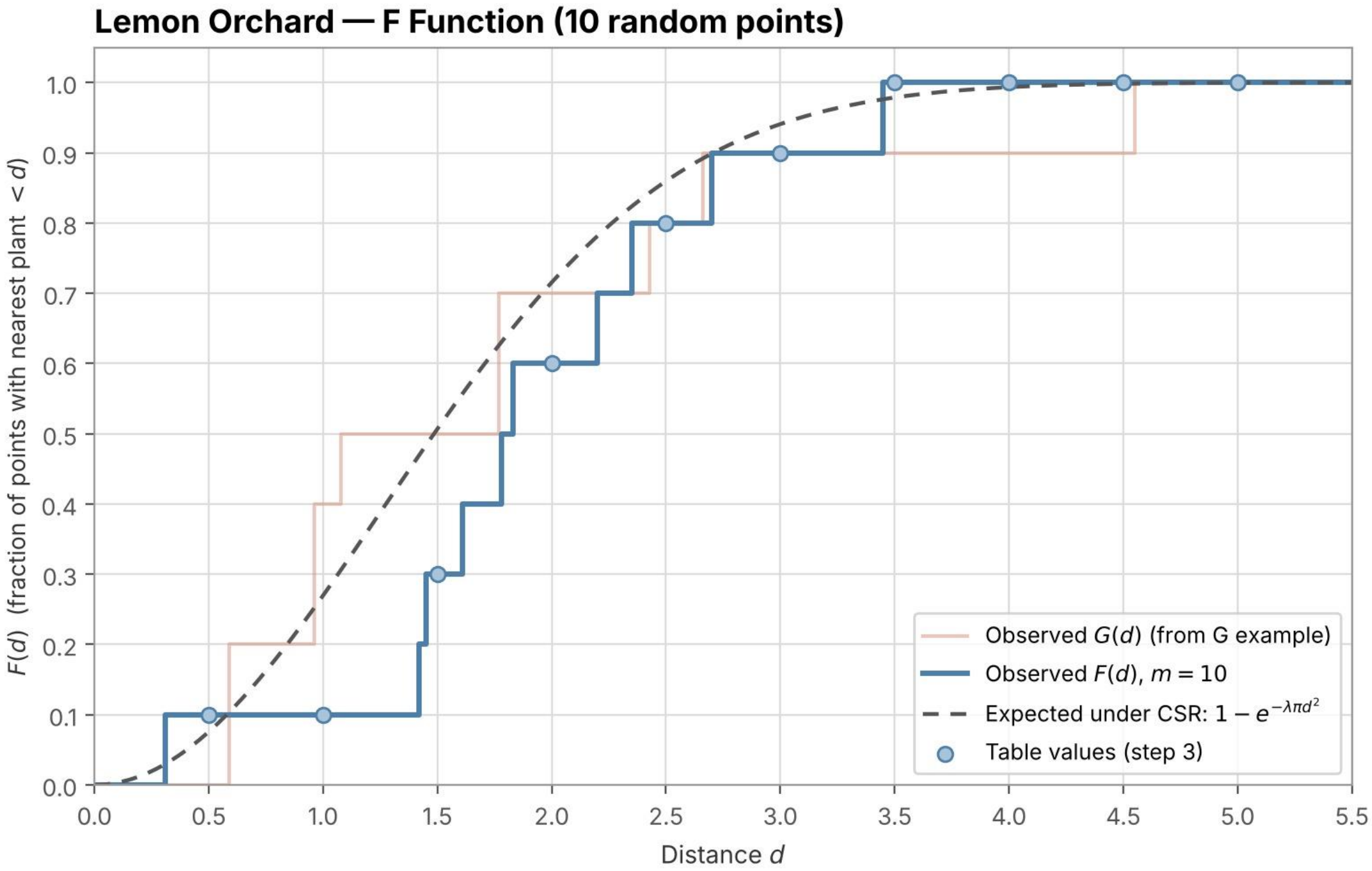


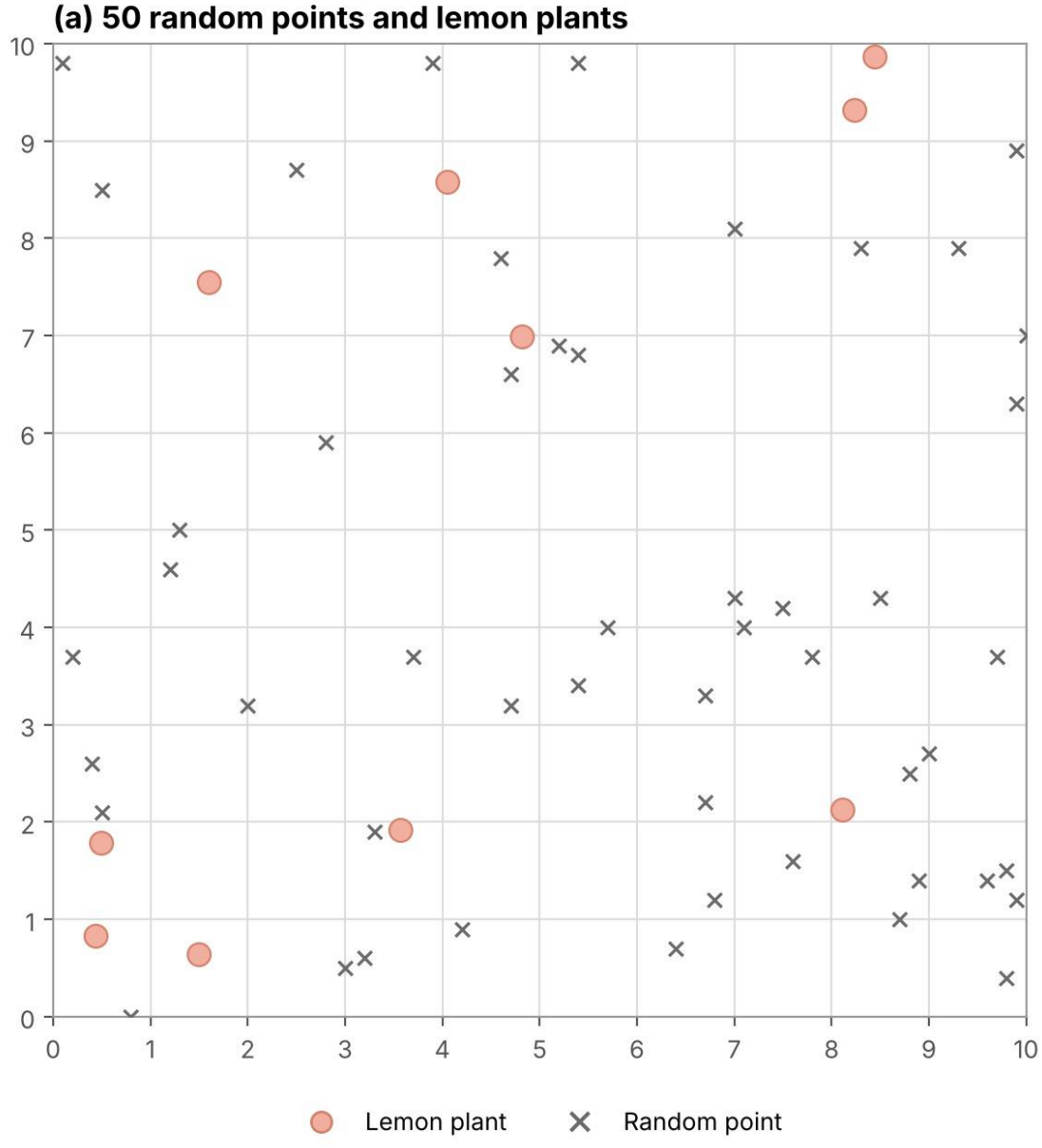
Grey crosses are random sample points P1–P10; dashed lines join each to its nearest lemon plant (distance labeled).

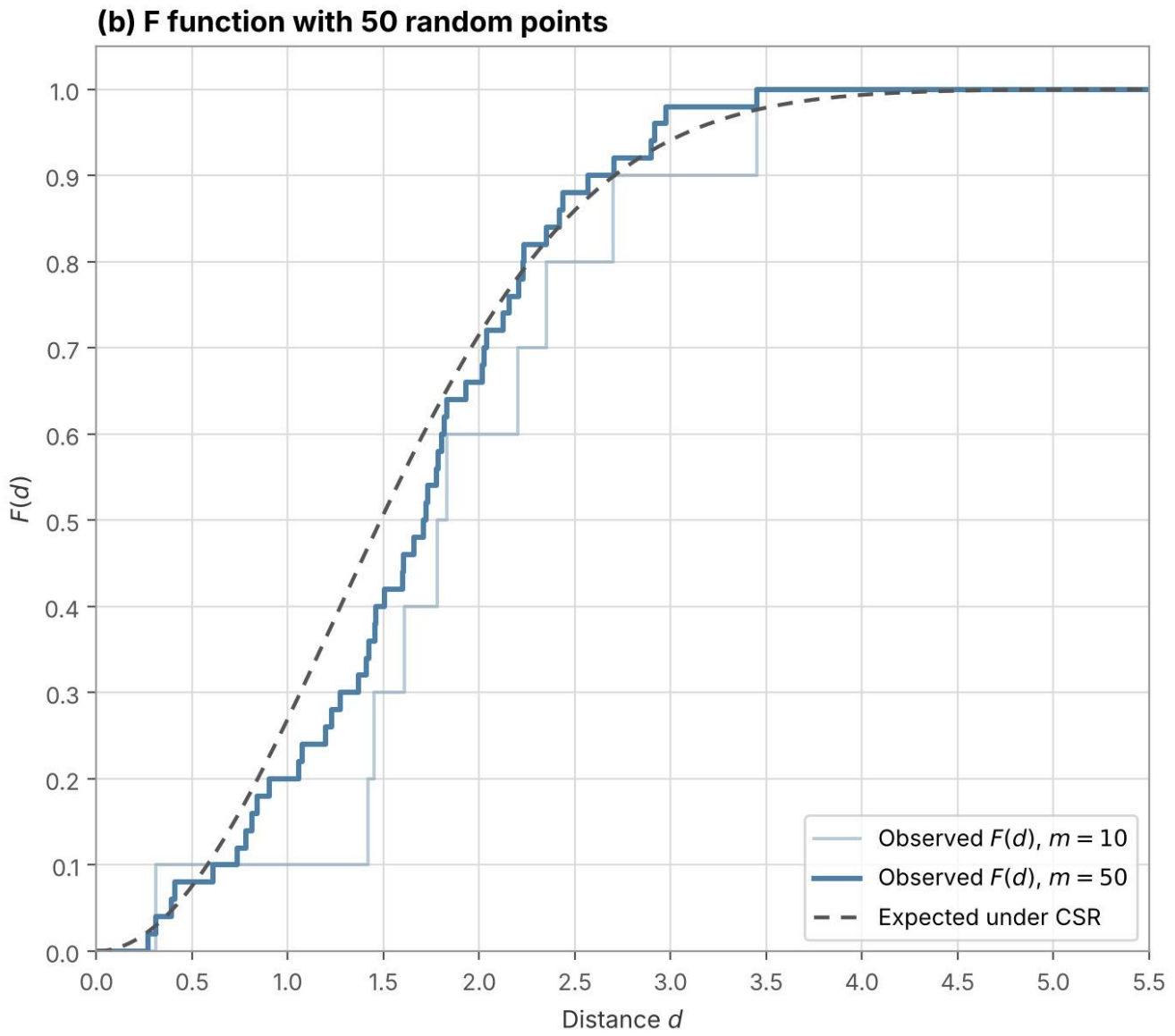
F FUNCTION

Examines the cumulative frequency of NN distances from a set of points randomly placed in the study area to the closest point in the pattern

1. Randomly place *m* points in the study area
2. Measure NN distance to each lemon
3. Record cumulative frequency of neighbors as distance increases


Lemon Orchard — F Function (10 random points)
F(d) (fraction of points with nearest plant < d)
Distance d
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
5.5
Observed G(d) (from G example)
Observed F(d), m = 10
Expected under CSR: 1 − e^{−λπd²}
Table values (step 3)


(a) 50 random points and lemon plants
Lemon plant
Random point


(b) F function with 50 random points
F(d)
Distance d
Observed F(d), m = 10
Observed F(d), m = 50
Expected under CSR

QUESTION

Can you see a common shortcoming of both the *G* and *F* functions?

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Can you see a common shortcoming of both the *G* and *F* functions?

ANSWER: Both statistics use only the nearest neighbor for each event or location in a pattern. There is a lot more information available if you consider more neighbors in the pattern.

TAKEAWAYS

- Spatial patterns can be described and assessed using intensity- and/or separation-based approaches
- Edge effects are often part of the design of research. The placement of edges can alter statistical results, and must be treated with care.
- Separation-based statistics have progressively evolved to include more information from the point pattern.